Shahzar Rizvi

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Education

Massachusetts Institute of Technology **Fall 2024 – Present**
Electrical Engineering and Computer Science (PhD)

University of California, Berkeley
Statistics (MA) **Spring 2024**
Computer Science (Honors, BA), **Math** (BA), **Statistics** (BA), **Logic** (minor) **Fall 2021**

Research

Research Assistant under Tamara Broderick **Fall 2024 – Present**
at Massachusetts Institute of Technology (Department of Electrical Engineering and Computer Sciences)
Prediction-powered inference for spatial settings
- Establish the modes of failure of prediction-powered inference in spatial settings where the data is not iid.
- Investigate, develop, and compare adjusted methods to allow for statistically valid prediction-powered inference in spatial settings.

Inconsistency of the number of components in finite mixture models
- Find and develop the appropriate theoretical tools to prove that the number of components in a misspecified finite mixture model diverge to infinity.
- Investigate the conditions under which the number of components diverge to infinity to generalize the inconsistency result to different model-fitting methods and distributional assumptions.

Research Assistant under Ryan Giordano **Fall 2023 – Summer 2024**
at University of California, Berkeley (Department of Statistics)
Sensitivity of Gaussian processes to the choice of prior
- Applied linear response perturbation to empirical Bayes to examine the sensitivity of statistical estimates (particularly in Gaussian processes) to prior selection.
- Developed and implemented in Python linear response perturbation methods for general Bayesian frameworks.

Research Assistant under Nathan Lo **Fall 2022 – Spring 2024**
at Stanford University (Division of Infectious Diseases and Geographic Medicine)
Statistical methods for identifying predominant modes of disease transmission.
- Developed estimation techniques for distinguishing between epidemics driven predominantly by environmental or person-to-person modes of transmission
- Analyzed the sensitivity of the estimation techniques to partial case observation and case reporting inaccuracies.

Research Assistant under Benjamin Nachman **Spring 2022 – Spring 2024**
at the Berkeley Institute for Data Science
Learning likelihood ratios with neural network classifiers
- Developed classification loss functions which allow for the estimation of likelihood ratios of two distributions by training classifiers to distinguish between their samples.
- Compared the performance of these different loss functions in estimating the likelihood ratio in classical, synthetic, and real data settings.

Publications

Shahzar Rizvi, Mariel Pettee, & Benjamin Nachman. Learning likelihood ratios with neural network classifiers. *Journal of High Energy Particle Physics*. (2024 February 20)

Shahzar Rizvi, Sophia Tan, Nila Cibu, Benjamin Singer, Andrew Azman, & Nathan Lo. Household and temporal clustering of cases within epidemic curves to determine mode of infectious disease transmission [in progress]. Stanford University, Department of Medicine, Division of Infectious Diseases and Geographic Medicine.

Shahzar Rizvi, Mariel Pettee, & Benjamin Nachman. Loss functionals for learning likelihood ratios. *NeurIPS Workshop on Machine Learning and the Physical Sciences*. (2023 December 15).

Sophia Tan, **Shahzar Rizvi**, Nila Cibu, Benjamin Singer, Andrew Azman, & Nathan Lo. Identifying routes of hepatitis E virus infection during outbreaks through model-driven estimation [abstract]. *Proceedings of the 2023 Annual Meeting of the American Society of Tropical Medicine and Hygiene*. Abstract #6752. (2023 October 18-22).

Posters

Loss Functionals for Learning Likelihood Ratios **Fall 2023**
NeurIPS Workshop on Machine Learning and the Physical Sciences

Identifying Routes of Hepatitis E Virus Infection During Outbreaks Through Model-Driven Estimation **Fall 2023**
2023 Annual Meeting of the American Society of Tropical Medicine and Hygiene

Stress-Testing the Grid-Based NPMLE **Fall 2020**
UC Berkeley Statistics Undergraduate Research Fair

A Bayesian Approach to Super-Resolution Microscopy **Spring 2020**
UC Berkeley Statistics Undergraduate Research Fair

Synthesis and Characterization of Bis(Cp^{4}methylpryazolyl)pyridine Ligands **Spring 2019**
Saegebarth Undergraduate Research Fair (UC Berkeley College of Chemistry)

Investigating Aggregative Growth Patterns of Colloidal Platinum Nanoparticles **Spring 2017**
Alivisatos Group Undergraduate Research Fair (UC Berkeley)

Teaching

Data 140: Probability for Data Science **Various**
Head Teaching Assistant (Fall 2020, Spring 2021, Fall 2021, Spring 2023, Fall 2023)

Data 88S: Probability and Mathematical Statistics for Data Science **Various**
Course Instructor (Fall 2022, Summer 2023)

CS 70: Discrete Mathematics and Probability Theory **Various**
Course Instructor (Summer 2021, Summer 2024)
Head Teaching Assistant (Summer 2020)
Teaching Assistant (Fall 2020)

Service

Chair of the Doctoral Student Group for the MIT EECS Visiting Committee **Spring 2026**
Chair of the group of doctoral students that wrote and presented a report on the doctoral student experience to the biyearly MIT EECS Visiting Committee.

Honors and Awards

Outstanding GSI Award **2022**
Selected from the top GSIs for outstanding work in teaching at UC Berkeley.

Phi Beta Kappa: Liberal Arts and Sciences Honors Society **Summer 2021 – Present**
Chosen from the top Letters & Science students to become a member of the Phi Beta Kappa honors society.

EECS Honors Program **Fall 2020 – Present**
Admitted into the EECS Honors program for academically talented undergraduates also involved in research.

Upsilon Pi Epsilon: Computer Science Honors Society **Fall 2019 – Present**
Chosen from the top third of computer science students to help serve the computer science community at UC Berkeley.

Tau Beta Pi: Engineering Honors Society **Fall 2018 – Present**
Publicity Chair (Spring 2019)
Chosen from the top fifth of engineering students to develop and serve the engineering community at UC Berkeley.

Saegebarth Undergraduate Research Grant **Summer 2019**
Received the grant to conduct research for the Rittle Lab at UC Berkeley.

NCTE Certificate for Superior Writing **2017**
One of 118 students in the nation to receive a Certificate for Superior Writing.